

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

Topic: Bus Admittance Matrix, Power Flow Analysis and Fault Analysis

Degree, Semester & Branch: V Semester B.E. EEE

Course Code & Title: EE8501 Power System Analysis

Name of the Faculty member: Mr. A. Arun Kumar

Name of the Innovative Practice: Modern Tool Usage)

Date & Duration: 03.10.2019 (50 Minutes)

ICT Tool Used: Use of Simulation Software

Description: Representation of various Power System components in simulation software was demonstrated using Power World Simulation Software. Formation of Y_{bus} matrix, Power Flow Analysis and Fault analysis were explained.

Goals (Learning Outcomes):

- The students will be able to construct Y_{bus} matrix using Simulation software.
- The students will be able to analyse the real and reactive power flow in different transmission lines using Power Flow Analysis.
- The students will be able to determine the fault current when fault is occurred at various location of the power system.

Use of appropriate methods:

Justification for Choosing the Activity for the topic:

Construction of Y_{bus} for a large power system through manual calculation is difficult. Power flow analysis is the important study in any power system research. Also, solving the power flow analysis and fault analysis in the classroom is a conventional method to understand. Thus, students are exposed with simulation software to understand the concept clearly. When the students do research in Power Systems, usage of software will really help them. That's why this modern tool usage is selected as one of the Innovative Practices.

Time Allotted for the Activity: 50 Minutes

Effective presentation

Details of the Implementation:

After teaching all the concepts related with formation of Y_{bus} matrix, Power flow analysis and fault analysis in the classroom, the students were taken Power System Simulation Laboratory. Total Strength is 55 and the available Desktop system is 71. The students are given with several problems which they have already in the classroom. Each students was given a Desktop system to do the simulation experiments.

A view of implementation of the Activity while the students are doing simulation.

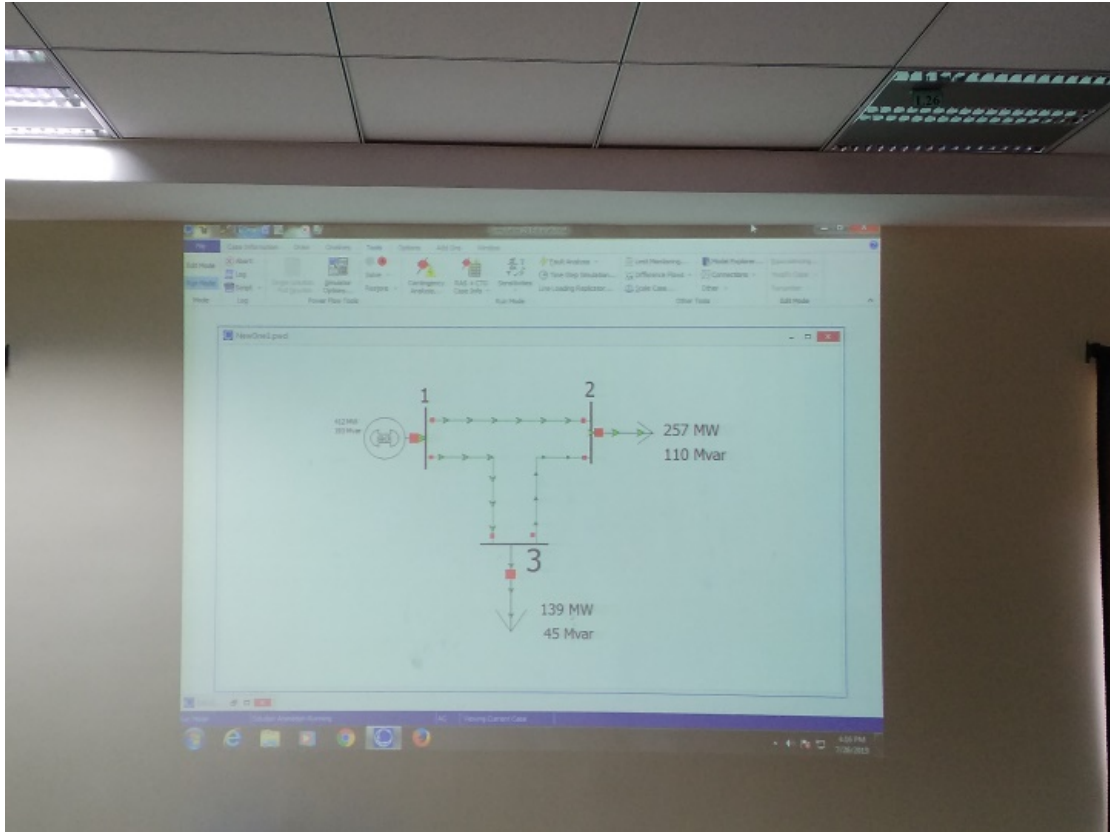


Figure 1



Figure 2



Figure 3



Figure 4

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when students were able to analyse the impact of line charging admittance on the Ybus matrix formation.
- ✓ The Students are able to classify different buses in Power Flow Analysis.
- ✓ The success of the activity was evaluated by asking the a 2 Mark question related with fault analysis has been asked in Internal Assessment Test III – **Out of 55 students, 35 answered correctly. Percentage is 66.63% (35/55).**

Reflective Critique:

1. Pre-implementation Reflection :

- **Benefits:**

- The students have analysed the importance of slack bus by simply connecting/disconnecting the slack bus generator.
The students were explained about the cause and effect of fault analysis effectively using the Power World Simulator.

- **Challenges:**

- Since the line charging admittance will have effect on only diagonal element. The student will find difficult to form Y_{bus} matrix.
- Since I handled for all the 55 students, the coordination for clarifying the doubts was time consuming.
- Learning a new software in a short span of time was difficult

2. Post-implementation Reflection :

- **Benefits:**

- The students have analysed the mathematical modelling of Power system using Bus admittance matrix.
- The same simulation, the students will do in VII Semester laboratory course “Power System Simulation Laboratory”. So they can easily do the experiments.
- The students calculated the fault current at different locations (near, middle and far away from the generator) Thereby variation of fault current is observed.
- They are also able to determine the most severe fault in Power system.

- **Challenges:**

- Students were not able to give the slack bus voltage correctly and line charging impedance was given at different location.
- Students found difficult to give the actual impedance of transmission line.

Changes required for future (if required):

- Instead of conducting for 50 minutes, the demonstration can be conducted for half a day.
- Students can be allowed to do the experiments on their own and can clarify their doubts.

- Guidance for using Free version can be given so that they will try to use the software when they are at home/hostel.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	3	3	2	1	1	1	-	-	-	-	-	1	2	2	1	1
CO2	3	3	2	1	2	1	-	-	-	-	-	1	2	2	1	1
CO3	3	2	2	1	2	2	-	-	-	-	-	1	2	2	1	1
CO4	3	3	1	2	1	2	-	-	-	-	-	1	2	2	1	1

CO1 Develop a suitable program for determining the transmission line parameters and its performance evaluation.

CO2 Simulate the power system to obtain the power flow solution using numerical iterative methods.

CO3 Perform fault analysis for a given power system under symmetrical and unsymmetrical fault.

CO4 Analyze the stability of the power system by simulation using appropriate techniques

References:

1. John J. Grainger, William D. Stevenson, Jr, 'Power System Analysis', Mc Graw Hill Education (India) Private Limited, New Delhi, 2015.
2. Kothari D.P. and Nagrath I.J., 'Power System Engineering', Tata McGraw-Hill Education, Second Edition, 2008.
3. Hadi Saadat, 'Power System Analysis', Tata McGraw Hill Education Pvt. Ltd., New Delhi, 21st reprint, 2010.
4. <https://www.powerworld.com/knowledge-base>